Consilience

Smart Contract Security Audit

Prepared by ShellBoxes

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Scope

The Consilience Contract in the Consilience Repository

Repo	Commit Hash
`https://github.com/tableturn/tt-white-contracts`	c301eefeee4bc9d111fb8757bb4f3a3ecf2e9c90

Files	MD5 Hash
marketplace/MarketplaceAccessFacet.sol	13df788e55901547cf1dc801fae14414
marketplace/MarketplaceInitFacet.sol	19560c6942dd3dd0d728bccc35fd69b6
marketplace/MarketplaceTokenHoldersFacet.sol	e5aef4fad710d83533cfdded3aa67ed1
marketplace/MarketplaceTopFacet.sol	977c3322cf604fe5a131ee026572269a
marketplace/lib/AMarketplaceFacet.sol	75d88531057c84c71e22aea296202071
marketplace/lib/IMarketplaceEvents.sol	34df7c70e1301c3c5e3ae9eb1785f9b2
marketplace/lib/LibMarketplace.sol	f9b8cbc23be4c531f4495c3bacf4dcc3
marketplace/lib/LibMarketplaceAccess.sol	a15801117aa01b591b57b025bbcb2f56

marketplace/lib/LibMarketplaceTokenHolders.sol	1c12a13359c47da543bdb2a0af3fa119
lib/LibAddressSet.sol	a5b0270111e90e7b3a4ee279b1ac6d73
lib/LibConstants.sol	436d7fb9ab3806a92f76109dbc91e392
lib/LibDiamond.sol	ea6a285ed94100d4db0f3df60a047ca2
lib/LibHelpers.sol	b8c18d7e9a0fd3b89d0ab08ff5f9d60e
lib/LibPaginate.sol	f631521eaafb54dee9a08b62fa70ac74
issuer/IssuerAccessFacet.sol	a21dc4bc5a51598297f96145c669bc00
issuer/IssuerFrontendFacet.sol	acbb8d719ce4b596e3ef0e025fc4a80c
issuer/IssuerInitFacet.sol	901e8af2f23483454979f4c40ecfd651
issuer/IssuerTopFacet.sol	9f472084d1479b4f425cde2124a59aa1
issuer/lib/AIssuerFacet.sol	936461a36b83eaccb40800a363540647
issuer/lib/IIssuerEvents.sol	94b82c5cf72559dced1c66f78b99098d
issuer/lib/LibIssuer.sol	d14dd750659ca2a183017d418f85f183
issuer/lib/LibIssuerAccess.sol	50f05d3bbcdb7d73d16231eba312a412
fast/FastAccessFacet.sol	7aa1770fec4958017c6c38a1063167ad
fast/FastFrontendFacet.sol	5654dc69424cfd79f460b7f0df644528
fast/FastHistoryFacet.sol	ed89b9179abb31c9685ccfe37eb32ffe
fast/FastInitFacet.sol	dd965459b83379b7eebf3d36814699e4
fast/FastTokenFacet.sol	850b81bd88e1163fdb936f5daa75de56
fast/FastTopFacet.sol	95545917c8a724d8624e920a449c05e3

fast/lib/AFastFacet.sol	d71dc5c444174bf26f86faa17e87bf47
fast/lib/IFast.sol	7e0241bf39fb8c1e690813fde31aa40c
fast/lib/IFastEvents.sol	eb023c68fc5b95d07bb92b3c321f9568
fast/lib/LibFast.sol	a6d67440b97122bd9d10d23d75a8b94c
fast/lib/LibFastAccess.sol	adb2c1b1dd5e08c0741d7b0bc2897d94
fast/lib/LibFastHistory.sol	daaf04ccd91967b7e8c41a86eef6267f
fast/lib/LibFastToken.sol	4aacbdb8141741a14474d5f6acaa9798

Re-Audit Scope

Repo	Commit Hash
`https://github.com/tableturn/tt-white-contracts`	f01917e8955c62a12310300b8afeb7990ce3111c

Files	MD5 Hash
marketplace/MarketplaceAccessFacet.sol	13df788e55901547cf1dc801fae14414
marketplace/MarketplaceInitFacet.sol	19560c6942dd3dd0d728bccc35fd69b6
marketplace/MarketplaceTokenHoldersFacet.sol	e5aef4fad710d83533cfdded3aa67ed1
marketplace/MarketplaceTopFacet.sol	977c3322cf604fe5a131ee026572269a
marketplace/lib/AMarketplaceFacet.sol	75d88531057c84c71e22aea296202071

marketplace/lib/IMarketplaceEvents.sol	34df7c70e1301c3c5e3ae9eb1785f9b2
marketplace/lib/LibMarketplace.sol	f9b8cbc23be4c531f4495c3bacf4dcc3
marketplace/lib/LibMarketplaceAccess.sol	a15801117aa01b591b57b025bbcb2f56
marketplace/lib/LibMarketplaceTokenHolders.sol	1c12a13359c47da543bdb2a0af3fa119
lib/LibAddressSet.sol	a5b0270111e90e7b3a4ee279b1ac6d73
lib/LibConstants.sol	436d7fb9ab3806a92f76109dbc91e392
lib/LibDiamond.sol	ea6a285ed94100d4db0f3df60a047ca2
lib/LibHelpers.sol	b8c18d7e9a0fd3b89d0ab08ff5f9d60e
lib/LibPaginate.sol	f631521eaafb54dee9a08b62fa70ac74
issuer/IssuerAccessFacet.sol	b1d2236361a10452130023ab2ca2310b
issuer/IssuerFrontendFacet.sol	acbb8d719ce4b596e3ef0e025fc4a80c
issuer/IssuerInitFacet.sol	bd6ea0d9be20be4ffa2b32a9ba73c8a9
issuer/IssuerTopFacet.sol	9f472084d1479b4f425cde2124a59aa1
issuer/lib/AIssuerFacet.sol	da2dea79c5398768e77e13a0af6a28cf
issuer/lib/IIssuerEvents.sol	94b82c5cf72559dced1c66f78b99098d
issuer/lib/LibIssuer.sol	d14dd750659ca2a183017d418f85f183
issuer/lib/LibIssuerAccess.sol	50f05d3bbcdb7d73d16231eba312a412
fast/FastAccessFacet.sol	7aa1770fec4958017c6c38a1063167ad
fast/FastFrontendFacet.sol	a841e15dc1dc24b2a5b0ced055fde0fe
fast/FastHistoryFacet.sol	ed89b9179abb31c9685ccfe37eb32ffe

fast/FastInitFacet.sol	6b385f7c8cb56ddecff869de85e9ce89
fast/FastTokenFacet.sol	d2bece36135fb014840eedf2c1482381
fast/FastTopFacet.sol	2db04d42eda896addcead77a1bf465db
fast/lib/AFastFacet.sol	690746367e84baa41287f6bb1bbbd5d0
fast/lib/IFast.sol	7e0241bf39fb8c1e690813fde31aa40c
fast/lib/IFastEvents.sol	6a9a8aaf827eeec5815ab08578cb4ed0
fast/lib/LibFast.sol	a101f9667384839ac112c1bff9e83831
fast/lib/LibFastAccess.sol	adb2c1b1dd5e08c0741d7b0bc2897d94
fast/lib/LibFastHistory.sol	daaf04ccd91967b7e8c41a86eef6267f
fast/lib/LibFastToken.sol	d1343335d99e7272a29941337fd4e9b4

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Contents

1 Introduction 9
- 1.1 About Consilience Group Limited . . . 9
- 1.2 Approach & Methodology . . . 9
  - 1.2.1 Risk Methodology . . . 10

2 Findings Overview 11
- 2.1 Summary . . . 11
- 2.2 Key Findings . . . 11

3 Finding Details 12
- A FastTokenFacet.sol . . . 12
  - A.1 Two Members Can Lock All The Fast's Funds [HIGH] . . . 12
  - A.2 The Issuer Can Retrieve Anyone's Tokens [MEDIUM] . . . 13
- B LibDiamond.sol . . . 15
  - B.1 All The Upgrades Should Be Performed Using DAO [MEDIUM] . . . 15
  - B.2 The Diamond Can End Up Without An Owner [LOW] . . . 16
- C IssuerAccessFacet.sol . . . 17
  - C.1 One Issuer Member May Remove Any Other Issuer Member [MEDIUM] 17
- D FastTokenFacet.sol . . . 19
  - D.1 The Issuer Member Can Burn The Fast's Tokens [MEDIUM] . . . 19

4 Attack Scenarioes 21
- 4.1 Issuer . . . 21
- 4.2 Fast . . . 21
- 4.3 Marketplace . . . 24

5 Best Practices 26
- BP.1 Unnecessary Initializations . . . 26
- BP.2 Unnecessary Argument . . . 26

6 Tests 28

7 Coverage 41

8 Static Analysis (Slither) 44

9 Conclusion 58

10 Disclaimer 59

1 Introduction

Consilience Group Limited engaged ShellBoxes to conduct a security assessment on the Consilience beginning on September 19th, 2022 and ending October 3rd, 2022. In this report, we detail our methodical approach to evaluate potential security issues associated with the implementation of smart contracts, by exposing possible semantic discrepancies between the smart contract code and design document, and by recommending additional ideas to optimize the existing code. Our findings indicate that the current version of smart contracts can still be enhanced further due to the presence of many security and performance concerns.

This document summarizes the findings of our audit.

1.1 About Consilience Group Limited

Consilience Ventures is the first start-up market network to fully align the interests of start-ups, venture capital investors and experienced and talented experts to help turn innovative ideas into high-growth businesses.

Issuer	Consilience Group Limited
Website	`https://consilience.vc/`
Type	Solidity Smart Contract
Audit Method	Whitebox

1.2 Approach & Methodology

ShellBoxes used a combination of manual and automated security testing to achieve a balance between efficiency, timeliness, practicability, and correctness within the audit's scope. While manual testing is advised for identifying problems in logic, procedure, and implementation, automated testing techniques help to expand the coverage of smart contracts and can quickly detect code that does not comply with security best practices.

1.2.1 Risk Methodology

Vulnerabilities or bugs identified by ShellBoxes are ranked using a risk assessment technique that considers both the LIKELIHOOD and IMPACT of a security incident. This framework is effective at conveying the features and consequences of technological vulnerabilities.

Its quantitative paradigm enables repeatable and precise measurement, while also revealing the underlying susceptibility characteristics that were used to calculate the Risk scores. A risk level will be assigned to each vulnerability on a scale of 5 to 1, with 5 indicating the greatest possibility or impact.

- Likelihood quantifies the probability of a certain vulnerability being discovered and exploited in the untamed.
- Impact quantifies the technical and economic costs of a successful attack.
- Severity indicates the risk's overall criticality.

Probability and impact are classified into three categories: H, M, and L, which correspond to high, medium, and low, respectively. Severity is determined by probability and impact and is categorized into four levels, namely Critical, High, Medium, and Low.

Impact	High	Medium	Low
High	Critical	High	Medium
Medium	High	Medium	Low
Low	Medium	Low	Low

Likelihood

2 Findings Overview

2.1 Summary

The following is a synopsis of our conclusions from our analysis of the Consilience implementation. During the first part of our audit, we examine the smart contract source code and run the codebase via a static code analyzer. The objective here is to find known coding problems statically and then manually check (reject or confirm) issues highlighted by the tool. Additionally, we check business logics, system processes, and DeFi-related components manually to identify potential hazards and/or defects.

2.2 Key Findings

In general, these smart contracts are well-designed and constructed, but their implementation might be improved by addressing the discovered flaws, which include , 1 high-severity, 4 medium-severity, 1 low-severity vulnerabilities.

Vulnerabilities	Severity	Status
A.1. Two Members Can Lock All The Fast's Funds	HIGH	Fixed
A.2. The Issuer Can Retrieve Anyone's Tokens	MEDIUM	Acknowledged
B.1. All The Upgrades Should Be Performed Using DAO	MEDIUM	Acknowledged
C.1. One Issuer Member May Remove Any Other Issuer Member	MEDIUM	Fixed
D.1. The Issuer Member Can Burn The Fast's Tokens	MEDIUM	Acknowledged
B.2. The Diamond Can End Up Without An Owner	LOW	Acknowledged

3 Finding Details

A FastTokenFacet.sol

A.1 Two Members Can Lock All The Fast's Funds [HIGH]

Description:

Each fast contains an attribute called transferCredits, the transfer credits are used to perform the transfer operation, whenever a fast is out of transfer credits, it becomes impossible for the member to transfer their tokens. However, this attribute depends on the fast itself and not the users, therefore the transfer credits are a shared resource between the fast members. Thus, two members can spend all the transfer credits of the fast resulting in a denial of service to all the members, locking their funds in the contract until the issuer member's intervention.

Code:

Listing 1: FastTokenFacet.sol

```
// If the funds are not moving from the zero address, decrease transfer credits.
if (p.from != address(0)) {
  s.transferCredits -= p.amount;
}
```

Risk Level:

Likelihood – 5
Impact - 4

Recommendation:

It is recommended to attach the transferCredits attribute to the member instead of the fast itself, or to limit the transfer credits that can be spent by one member of the fast.

Status - Fixed

The Consilience team has fixed the issue by removing the transfer credits functionality.

A.2 The Issuer Can Retrieve Anyone's Tokens [MEDIUM]

Description:

The retrieveDeadTokens function allows the issuer to transfer all the tokens of any account to the reserve. This represents a significant centralization risk where the issuer has too much control over all the members' accounts.

Code:

Listing 2: FastTokenFacet.sol

```
function retrieveDeadTokens(address holder)
    external
    onlyIssuerMember {
  // Cache how many tokens the holder has.
  uint256 amount = balanceOf(holder);
  // Note: The amount **can** be zero in this function.

  // Grab a pointer to the token storage.
  LibFastToken.Data storage s = LibFastToken.data();

  // These should only run if the amount is zero, as they result in a no-op.
  if (amount > 0) {
    // Set the holder balance to zero.
    s.balances[holder] = 0;
    // Increment the reserve's balance.
    s.balances[address(0)] += amount;
    // The tokens aren't in circulation anymore - decrease total supply.
    s.totalSupply -= amount;
```

```
    }

    // Since the holder's account is now empty, make sure to keep track of it both
    // in this FAST and in the marketplace.
    s.tokenHolders.remove(holder, true);
    MarketplaceTokenHoldersFacet(LibFast.data().marketplace).fastBalanceChanged(holder, 0);

    // This operation can be seen as a regular transfer between holder and reserve. Emit.
    emit Transfer(holder, address(0), amount);

    // If amount wasn't zero, total supply and reserve balance have changed - emit.
    if (amount > 0) {
      FastFrontendFacet(address(this)).emitDetailsChanged();
    }
  }
```

Risk Level:

Likelihood – 2
Impact - 5

Recommendation:

Consider using a multisig wallet or a DAO as the address of the issuer in order to avoid centralization risks and include multiple parties in the decision-making.

Status - Acknowledged

The Consilience team has acknowledged the issue, stating that the functionality cannot be changed due to regulatory restrictions under the Financial Conduct Authority (FCA) requirements in article SUP 10A.10 Customer-dealing functions.

B LibDiamond.sol

B.1 All The Upgrades Should Be Performed Using DAO [MEDIUM]

Description:

The contracts implement Tokenization as a Service using the diamond pattern, this implementation offers the upgradability of functionalties along with other advantages. However, by using this approach the contract owner can upgrade the contract to implement any logic, this represents a centralization risk knowing that the contract owner can change the logic of the code anytime which can cause unexpected behaviors to the users.

Code:

Listing 3: LibDiamond.sol

```
function diamondCut(
    IDiamondCut.FacetCut[] memory _diamondCut,
    address _init,
    bytes memory _calldata
) internal {
    for (uint256 facetIndex; facetIndex < _diamondCut.length; facetIndex ++) {
        IDiamondCut.FacetCutAction action = _diamondCut[facetIndex].action;
        if (action == IDiamondCut.FacetCutAction.Add) {
            addFunctions(_diamondCut[facetIndex].facetAddress, _diamondCut[facetIndex].functionSelectors);
        } else if (action == IDiamondCut.FacetCutAction.Replace) {
            replaceFunctions(_diamondCut[facetIndex].facetAddress, _diamondCut[facetIndex].functionSelectors);
        } else if (action == IDiamondCut.FacetCutAction.Remove) {
            removeFunctions(_diamondCut[facetIndex].facetAddress, _diamondCut[facetIndex].functionSelectors);
```

```
        } else {
            revert("LibDiamondCut: Incorrect FacetCutAction");
        }
    }
    emit DiamondCut(_diamondCut, _init, _calldata);
    initializeDiamondCut(_init, _calldata);
}
```

Risk Level:

Likelihood – 1
Impact - 4

Recommendation:

Consider using a DAO as the contract owner in order to include the community in the decision of upgrades.

Status - Acknowledged

The Consilience team has acknowledged the issue, stating that the functionality cannot be changed due to regulatory restrictions under the Financial Conduct Authority (FCA) requirements in article SUP 10A.10 Customer-dealing functions.

B.2 The Diamond Can End Up Without An Owner [LOW]

Description:

The setContractOwner allows the contract owner to delegate the ownership to another address. However, if the contract owner enters a wrong address or the address(0) as the _newOwner, the contract will have no owner, which results in a Denial Of Service in the privileged actions.

Code:

Listing 4: LibDiamond.sol

```
function setContractOwner(address _newOwner) internal {
    DiamondStorage storage ds = diamondStorage();
    address previousOwner = ds.contractOwner;
    ds.contractOwner = _newOwner;
    emit OwnershipTransferred(previousOwner, _newOwner);
}
```

Risk Level:

Likelihood – 1
Impact - 3

Recommendation:

Consider requiring the _newOwner to be different from the address(0), and it is recommended to implement a process where the contract owner sets an address as a new owner candidate, then this address can only be the contract owner if it calls the contract to accept the ownership.

Status - Acknowledged

The Consilience team has acknowledged the issue, stating that they want ownership to be transferable to the address(0) as a proof of future immutability.

C IssuerAccessFacet.sol

C.1 One Issuer Member May Remove Any Other Issuer Member [MEDIUM]

Description:

The removeMember function can be called by any member. Allowing a member to remove another one with the same privilege can harm the logic of the contract. For in instance, a

new issuer member will be able to remove all the other members.

Code:

Listing 5: IssuerAccessFacet.sol

```
function removeMember(address member)
  external override
  onlyMember(msg.sender) {
// No suicide allowed.
if (msg.sender == member) {
  revert ICustomErrors.CannotSelfRemove(msg.sender);
}
// Remove the member from the set.
LibIssuerAccess.data().memberSet.remove(member, false);
// Emit!
emit MemberRemoved(member);
}
```

Risk Level:

Likelihood – 2
Impact - 4

Recommendation:

Consider restricting the functionality to one master issuer member that will have a higher privilege over the other members.

Status - Fixed

The Consilience team has fixed the issue by only allowing the diamond owner to remove the issuer members.

D FastTokenFacet.sol

D.1 The Issuer Member Can Burn The Fast's Tokens [MEDIUM]

Description:

Anyone of the issuer members is able to burn tokens from the reserve, this action can represent a significant centralization risk where the issuer members have too much control over the fast's funds.

Code:

Listing 6: FastTokenFacet.sol

```
function burn(uint256 amount, string calldata ref)
  external
  onlyIssuerMember {
LibFastToken.Data storage s = LibFastToken.data();

if (FastTopFacet(address(this)).hasFixedSupply()) {
  revert ICustomErrors.RequiresContinuousSupply();
}

// Remove the minted amount from the zero address.
s.balances[address(0)] -= amount;

// Keep track of the minting operation.
FastHistoryFacet(address(this)).burnt(amount, ref);

// Emit!
FastFrontendFacet(address(this)).emitDetailsChanged();
emit Burnt(amount, ref);
}
```

Risk Level:

Likelihood – 2
Impact - 4

Recommendation:

Consider adding some restrictions to the burn functionality to limit the power of the issuer members over the fasts.

Status - Acknowledged

The Consilience team has acknowledged the issue, stating that the functionality cannot be changed due to regulatory restrictions under the Financial Conduct Authority (FCA) requirements in article SUP 10A.10 Customer-dealing functions.

4 Attack Scenarioes

4.1 Issuer

Registering a fast without being an issuer member

X FAILED

Protection by the onlyMember modifier

Adding/Removing Issuer members without being an issuer member

X FAILED

Protection by the onlyMember modifier

Adding/Removing Issuer members being an issuer member

✔ PASSED

The onlyMember modifier is not enough as all the members have the same privilege over each other

Adding/Removing a fast governor directly from the fastGovernorships mapping in the Issuer diamond without calling the Fast

X FAILED

Having in place a check that makes sure governorAddedToFast and governorRemovedFromFast can only be called by a registered Fast

4.2 Fast

Minting an amount without being an issuer member

X FAILED

Protection by the onlyIssuerMember modifier

- Minting more than one time in a fast that has a fixed supply

  X FAILED

  Having in place a check that makes sure the mint can only be performed one time in a fixed supply fast.

- Burning an amount without being an issuer member

  X FAILED

  Protection by the onlyIssuerMember modifier

- Burning in a fast that has a fixed supply

  X FAILED

  Having a check in place that makes sure the burn can only be performed in a continuous supply fast.

- Causing an overflow/underflow and generating unexpected results

  X FAILED

  Built in overflow protection in solidity 0.8.* versions

- Approve Race Condition

  X FAILED

  The allowance is incremented instead of being set to amount

- Transferring tokens to a non-fast-member in a private fast

  X FAILED

  The onlyTokenHolder modifier requires the from and to to be fast members when the fast is private

- Transferring tokens to a non-marketplace-member in a semi public fast

  X FAILED

  The onlyTokenHolder modifier requires the to to be a marketplace member when the fast is semi public

- Transferring tokens from the reserve without being a fast governor

  X FAILED

  The transfer function verifies the spender to be a fast governor whenever the from is the reserve address

- Performing a transfer without having enough transfer credits

  X FAILED

  The transfer call fails due to underflow protection

- Changing a fast from semi public to private

  X FAILED

  The setIsSemiPublic function prevents the change when a fast is semi public

- Adding/Removing governors without being an issuer member

  X FAILED

  Protection by the onlyIssuerMember modifier

- Adding/Removing fast members without being a governor

  X FAILED

  Protection by the onlyGovernor modifier

Adding a non-marketplace-member as a fast member

X FAILED

Protection by onlyMarketplaceMember modifier

Spending all the transfer credits by one member

X FAILED

The member is limited with his token balance

Spending all the transfer credits by two members

✔ PASSED

Two members can spend all the transfer credits by performing multiple transfer calls between each other

4.3 Marketplace

Adding/Removing marketplace members without being an issuer member

X FAILED

Protection by the onlyIssuerMember modifier

Adding/Removing a fast member directly from the fastMemberships mapping in the Marketplace diamond without calling the Fast

X FAILED

Having in place a check that makes sure memberAddedToFast and memberRemovedFromFast can only be called by a registered Fast

- Manipulating the fastHoldings mapping in the Marketplace diamond without calling the Fast

  - X FAILED

  - Having in place a check that makes sure fastBalanceChanged can only be called by a registered Fast

Conclusion:

The results of the attack scenarios are mentionned in the Finding Details section, the contracts were also tested for reentrancy attacks, front running attacks, Block Timestamp manipulation... , and no issues were found.

5 Best Practices

BP.1 Unnecessary Initializations

Description:

When a variable is declared in solidity, it gets initialized with its type's default value. Thus, there is no need to initialize a variable with the default value.

Code:

Listing 7: FastInitFacet.sol

```
tokenData.totalSupply = 0;
// Initialize other internal stuff.
tokenData.transferCredits = 0;
```

BP.2 Unnecessary Argument

Description:

The setIsSemiPublic function is used to change the fast's type. However, the fast's type can only be changed from private to semi-public, therefore the flag argument is unnecessary since the isSemiPublic attribute can be set directly to true in order to change the fast's type from private to semi-public.

Code:

Listing 8: FastTopFacet.sol

```
function setIsSemiPublic(bool flag)
  external
  onlyIssuerMember { // can be set to true
// Someone is trying to toggle back to private?... No can do!
if (this.isSemiPublic()) {
```

```
  revert ICustomErrors.UnsupportedOperation();
}
LibFast.data().isSemiPublic = flag;
// Emit!
FastFrontendFacet(address(this)).emitDetailsChanged();
}
```

6 Tests

Results:

```
FastAccessFacet
  IHasGovernors implementation
    isGovernor
      returns true when the address is a governor
      returns false when the address is not a governor
    governorCount
      returns the current count of governors
    paginateGovernors
      returns the cursor to the next page
      does not crash when overflowing and returns the correct cursor
      returns the governors in the order they were added
    addGovernor
      requires Issuer membership (anonymous) (42ms)
      requires Issuer membership (governor)
      delegates to the Issuer for permission checking
      requires that the address is an Marketplace member
      requires that the address is not a governor yet (52ms)
      adds the given address as a governor (49ms)
      calls FastFrontendFacet.emitDetailsChanged
      emits a GovernorAdded event
    removeGovernor
      requires Issuer membership (anonymous)
      requires Issuer membership (governor)
      delegates to the Issuer for permission checking
      requires that the address is an existing governor
      removes the given address as a governor (51ms)
      calls FastFrontendFacet.emitDetailsChanged
      emits a GovernorRemoved event
  IHasMembers
    isMember
```

```
      returns true when the address is a member
      returns false when the address is not a member
    memberCount
      returns the current count of members
    paginateMembers
      returns the cursor to the next page
      does not crash when overflowing and returns the correct cursor
      returns the members in the order they were added
    addMember
      requires governance (anonymous)
      requires governance (Issuer governor)
      requires that the address is an Marketplace member
      requires that the address is not a member yet (57ms)
      adds the given address as a member (43ms)
      delegates to the Marketplace contract to signal the membership
         ↪ addition
      calls FastFrontendFacet.emitDetailsChanged
      emits a MemberAdded event
    removeMember
      requires governance (anonymous)
      requires governance (Issuer governor)
      requires that the address is an existing member
      removes the given address as a member
      delegates to the token contract (40ms)
      delegates to the Marketplace contract to signal the membership
         ↪ addition
      calls FastFrontendFacet.emitDetailsChanged (41ms)
      emits a MemberRemoved event
  flags
    is accurate when all flags set (83ms)
    is accurate when only isGovernor is set (39ms)
    is accurate when only isMember is set (64ms)
    is accurate when no flags are set (42ms)
```

```
FastFrontendFacet
  emitDetailsChanged
     requires that the caller is the diamond
     emits a DetailsChanged event with all the correct information
  details
     returns a populated details struct
  detailedMember
     returns a MemberDetails struct with the correct information
  detailedGovernor
     returns a GovernorDetails struct with the correct information
  paginateDetailedMembers
     returns member details with next cursor
     handles an offset index cursor
  paginateDetailedGovernors
     returns governor details with next cursor

FastHistoryFacet
  minted
     requires that the caller is the token (anonymous)
     requires that the caller is the token (governor)
    as the diamond
       adds an entry to the supply proof list
  burnt
     requires that the caller is the diamond (anonymous)
     requires that the caller is the diamond (governor)
    as the diamond
       adds an entry to the supply proof list
  supplyProofCount
     counts how many supply proofs have been stored
  paginateSupplyProofs
     returns the cursor to the next page
     does not crash when overflowing and returns the correct cursor
     returns the supply proofs in the order they were added
  transfered
```

```
    requires that the caller is the token (anonymous)
    requires that the caller is the token (governor)
    adds an entry to the transfer proof list (76ms)
  transferProofCount
    counts how many transfer proofs have been stored (88ms)
  paginateTransferProofs
    returns the cursor to the next page
    does not crash when overflowing and returns the correct cursor
    returns the transfer proofs in the order they were added
  paginateTransferProofsByInvolvee
    returns the cursor to the next page
    does not crash when overflowing and returns the correct cursor (
      ↪ bob)
    does not crash when overflowing and returns the correct cursor (
      ↪ john)
   - counts the proofs regardless of the involvement (sender and
      ↪ recipient)
    categorizes the proofs for the senders
  paginateTransferProofIndicesByInvolvee
    returns a paginated list of addresses and cursor
  transferProofByInvolveeCount
    returns the count of the transfer proofs for a given address

FastTokenFacet
  initialize
    keeps track of the ERC20 parameters and extra ones (40ms)
  name
    returns the name
  symbol
    returns the symbol
  decimals
    returns the decimals
  totalSupply
    returns the total supply
```

```
transferCredits
   returns the remaining transfer credits
mint
   requires Issuer membership (anonymous)
   requires Issuer membership (member)
   requires Issuer membership (governor)
   delegates to the history contract (48ms)
   adds the minted tokens to the zero address (59ms)
   does not impact total supply (61ms)
   emits a Minted event (53ms)
  - delegates to the frontend facet
  with fixed supply
     is allowed only once (73ms)
  with continuous supply
     is allowed more than once (99ms)
burn
   requires Issuer membership (anonymous)
   requires Issuer membership (member)
   requires Issuer membership (governor)
   requires that the supply is continuous
   requires that the zero address has enough funds
   removes tokens from the zero address (56ms)
   does not impact total supply (48ms)
   delegates to the history contract
   emits a Burnt event (38ms)
  - delegates to the frontend facet
retrieveDeadTokens
   requires Issuer membership
   still emits a Transfer event if the balance was already zero
   sets the holder balance to zero while increasing the reserve
      ↪ balance (66ms)
   decreases the total supply by the amount (47ms)
   removes the holder from the FAST token holder list (43ms)
   calls the marketplace to stop tracking this token holder for this
```

```
      ↪ FAST
    emits a Transfer event between the holder and the reserve
    delegates to the Frontend facet for a global event emission (40ms)
addTransferCredits
    requires Issuer membership (anonymous)
    requires Issuer membership (member)
    requires Issuer membership (governor)
    accumulates the credits to the existing transfer credits (103ms)
    emits a TransferCreditsAdded event
  - delegates to the frontend facet
drainTransferCredits
    requires Issuer membership (anonymous)
    requires Issuer membership (member)
    requires Issuer membership (governor)
    sets the credit amount to zero (61ms)
    emits a TransferCreditsDrained event (38ms)
ERC20
  balanceOf
      returns the amount of tokens at a given address
  transfer
      delegates to the internal performTransfer method (115ms)
  transferWithRef
      delegates to the internal performTransfer method (78ms)
  allowance
      returns the allowance for a given member (44ms)
      follows value at zero address for governors (76ms)
  approve
      delegates to the internal performApproval method
      requires FAST membership
      adds an allowance with the correct parameters (117ms)
      functions properly when given a zero amount (97ms)
      stacks up new allowances (105ms)
      keeps track of given allowances (47ms)
      keeps track of received allowances (48ms)
```

```
    emits an Approval event
disapprove
    delegates to the internal Disapproval method
    subtracts from the existing allowance
    emits a Disapproval event
   when the allowance remains positive after the operation
      removes the spender received allowance when it reaches zero
      removes the original given allowance when it reaches zero
   when the allowance reaches zero
      removes the spender received allowance when it reaches zero
      removes the original given allowance when it reaches zero
transferFrom
    delegates to the internal performTransfer method (79ms)
transferFromWithRef
    delegates to the internal performTransfer method (85ms)
    decreases the transfer credits when not transacting from the
        ↪ zero address (128ms)
    requires that the sender and recipient are different
    requires sufficient funds (83ms)
    requires sufficient transfer credits (129ms)
    transfers from / to the given wallet address (132ms)
    delegates to the history contract (76ms)
    delegates to the MarketplaceTokenHoldersFacet contract (76ms)
    updates who holds this token (87ms)
    decreases total supply when transferring to the zero address
        ↪ (112ms)
    emits a IERC20.Transfer event (90ms)
    requires that there is enough allowance (52ms)
    allows non-members to transact on behalf of members (189ms)
    increases total supply when transferring from the zero address
        ↪ (96ms)
    requires that zero address can only be spent from as a governor
        ↪ (Issuer member) (48ms)
    requires that zero address can only be spent from as a governor
```

```
        ↪ (member) (43ms)
     requires that zero address can only be spent from as a governor
        ↪ (anonymous) (120ms)
     allows governors to transfer from the zero address (151ms)
     does not require transfer credits when drawing from the zero
        ↪ address (126ms)
     does not impact transfer credits when drawing from the zero
        ↪ address (165ms)
    when member deactivated
      requires active member when transferring from address (at the
         ↪ Marketplace level) (55ms)
      allows transfer to a deactived member (at the Marketplace
         ↪ level) (155ms)
    when semi-public
      requires sender membership (Marketplace membership)
      requires recipient membership (Marketplace membership) (40ms)
      allows marketplace members to transact (169ms)
    when private
      requires sender membership (FAST member)
      requires recipient membership (FAST member) (54ms)
givenAllowanceCount
   returns the count of allowancesByOwner
paginateAllowancesByOwner
   returns the list of addresses to which the caller gave allowances
      ↪ to
   does not list addresses from which the caller has received
      ↪ allowances
receivedAllowanceCount
   returns the count of allowancesBySpender
paginateAllowancesBySpender
   returns the list of addresses to which the caller gave allowances
      ↪ to
   does not list addresses to which the caller has given allowances
beforeRemovingMember
```

```
    cannot be called directly
   when successful
     reverts if the member to remove still has a positive balance
        ↪ (125ms)
     sets allowances to / from the removed members to zero (108ms)
     removes given and received allowances (74ms)
     emits a Disapproval event as many times as it removed allowance
        ↪ (68ms)

FastTopFacet
  issuerAddress
    returns the Issuer address
  marketplaceAddress
    returns the marketplace address
  isSemiPublic
    returns the FAST semi-public parameter
  hasFixedSupply
    returns the FAST fixed supply parameter
  setIsSemiPublic
    requires Issuer membership for the sender
    delegates to the Issuer for permission check (39ms)
    prevents changing from semi-public to closed (73ms)
    sets the required flag on the FAST (47ms)
    delegates to FastFrontendFacet.emitDetailsChanged (41ms)

IssuerAccessFacet
  IHasMembers
   isMember
     returns true when the candidate is a member
     returns false when the candidate is not a member
   memberCount
     correctly counts members
   paginateMembers
     returns pages of members (54ms)
```

```
    addMember
      requires that the sender is a member
      adds the member to the list
      does not add the same member twice
      emits a MemberAdded event
    removeMember
      requires that the sender is a member
      requires that the user is not removing themselves
      removes the member from the list
      reverts if the member is not in the list
      emits a MemberRemoved event
  governorAddedToFast
    requires the caller to be a registered FAST
    adds the given member to the FAST governorship tracking data
      ↪ structure (66ms)
    emits GovernorshipAdded event (55ms)
  governorRemovedFromFast
    requires the caller to be a registered FAST
    adds the given member to the FAST governorship tracking data
      ↪ structure (92ms)
    emits GovernorshipRemoved event (77ms)
  paginateGovernorships
    given an address, returns the list of FASTs that it is a governor
      ↪ of

IssuerFrontendFacet
  paginateDetailedFasts
    returns a paginated list of detailed FAST details (45ms)

IssuerInitFacet
  initialize
    requires that it is not initialized
    set various storage versions
    registers supported interfaces (50ms)
```

```
    adds the given address to the member list
    emits a MemberAdded event

IssuerTopFacet
  FAST management
    isFastRegistered
      returns false when the FAST symbol is unknown
      returns true when the FAST symbol is registered
    fastBySymbol
      returns the zero address when the FAST symbol is unknown
      returns the FAST address when the FAST symbol is registered
    registerFast
      requires Issuer membership
      reverts if trying to add a FAST with an already existing symbol
        ↪ (50ms)
      adds the registry address to the list of registries
      keeps track of the symbol
      emits a FastRegistered event
    fastCount
      returns the FAST count
    paginateFasts
      returns pages of FASTs

MarketplaceAccessFacet
  IHasMembers
    isMember
      returns true when the address is a member
      returns false when the address is not a member
    memberCount
      returns the current count of members
    paginateMembers
      returns the cursor to the next page
      does not crash when overflowing and returns the correct cursor
      returns the governors in the order they were added
```

```
  addMember
    requires Issuer membership (anonymous)
    delegates to the Issuer for permission (42ms)
    requires that the address is not a member yet (76ms)
    adds the given address as a member (59ms)
    emits a MemberAdded event
  removeMember
    requires Issuer membership (anonymous)
    delegates to the Issuer for permission
    requires that the address is an existing member - calls
       ↪ LibAddressSet
    requires that the given member has no FAST memberships (85ms)
    removes the given address as a member (59ms)
    emits a MemberRemoved event (40ms)
fastMemberships
   returns an array of FASTs a given user belongs to along with a
      ↪ cursor
   does not return FASTs the given user does not belong to
memberAddedToFast
   requires the caller to be a registered FAST
   adds the given member to the FAST membership tracking data
      ↪ structure (62ms)
memberRemovedFromFast
   requires the caller to be a registered FAST
   removes the FAST contract from the list of Fast members (95ms)
isMemberActive
   returns true when a member is active
   returns false when a member is deactived
deactivateMember
   requires the caller to be an Issuer member
   requires the member to deactivate is an Marketplace member
   adds the FAST member to the list of deactivated members (65ms)
   emits a MemberDeactivated event (49ms)
   requires that a given member is not already deactivated (94ms)
```

```
  activateMember
    requires the caller to be an Issuer member
    requires the member to activate is an Marketplace member
    removes the FAST member from the list of deactivated members (70ms
      ↪ )
    emits a MemberActivated event (50ms)
    requires that a given member is currently deactivated (79ms)

MarketplaceInitFacet
  initialize
    requires that it is not initialized (147ms)
    set various storage versions
    registers supported interfaces (92ms)
    stores the given Issuer address

MarketplaceTokenHoldersFacet
  holdingsUpdated
    reverts if not called by a FAST contract
    returns a list of FASTs that an account holds (64ms)
    removes the FAST holding if account balance drops to 0 (93ms)
    does not track the zero address (46ms)
  holdings
    returns a list of FASTs a account holds (60ms)

MarketplaceTopFacet
  issuerAddress
    returns the Issuer address


270 passing (2m)
4 pending
```

7 Coverage

Results:

```
File | % Stmts | % Branch | % Funcs | % Lines |Uncovered Lines |

 fast/ | 100 | 91.07 | 100 | 98.68 | |
  FastAccessFacet.sol | 100 | 100 | 100 | 100 | |
  FastFrontendFacet.sol | 100 | 100 | 100 | 100 | |
  FastHistoryFacet.sol | 100 | 100 | 100 | 100 | |
  FastInitFacet.sol | 100 | 50 | 100 | 96 | 47 |
  FastTokenFacet.sol | 100 | 90.24 | 100 | 98.43 | 283,428 |
  FastTopFacet.sol | 100 | 100 | 100 | 100 | |
 fast/lib/ | 88.89 | 77.78 | 92.31 | 83.87 | |
  AFastFacet.sol | 88.89 | 77.78 | 88.89 | 81.48 | 36,37,39,45,99 |
  IFast.sol | 100 | 100 | 100 | 100 | |
  IFastEvents.sol | 100 | 100 | 100 | 100 | |
  LibFast.sol | 100 | 100 | 100 | 100 | |
  LibFastAccess.sol | 100 | 100 | 100 | 100 | |
  LibFastHistory.sol | 100 | 100 | 100 | 100 | |
  LibFastToken.sol | 100 | 100 | 100 | 100 | |
 interfaces/ | 100 | 100 | 100 | 100 | |
  ICustomErrors.sol | 100 | 100 | 100 | 100 | |
  IDiamondCut.sol | 100 | 100 | 100 | 100 | |
  IDiamondLoupe.sol | 100 | 100 | 100 | 100 | |
  IERC1404.sol | 100 | 100 | 100 | 100 | |
  IERC165.sol | 100 | 100 | 100 | 100 | |
  IERC173.sol | 100 | 100 | 100 | 100 | |
  IERC20.sol | 100 | 100 | 100 | 100 | |
  IHasActiveMembers.sol | 100 | 100 | 100 | 100 | |
  IHasGovernors.sol | 100 | 100 | 100 | 100 | |
  IHasMembers.sol | 100 | 100 | 100 | 100 | |
 issuer/ | 100 | 88.89 | 100 | 97.96 | |
```

IssuerAccessFacet.sol | 100 | 90 | 100 | 94.44 | 108 |
IssuerFrontendFacet.sol | 100 | 100 | 100 | 100 | |
IssuerInitFacet.sol | 100 | 75 | 100 | 100 | |
IssuerTopFacet.sol | 100 | 100 | 100 | 100 | |
issuer/lib/ | 66.67 | 50 | 80 | 63.64 | |
AIssuerFacet.sol | 66.67 | 50 | 66.67 | 55.56 | 24,25,27,33 |
IIssuerEvents.sol | 100 | 100 | 100 | 100 | |
LibIssuer.sol | 100 | 100 | 100 | 100 | |
LibIssuerAccess.sol | 100 | 100 | 100 | 100 | |
lib/ | 29.13 | 24.29 | 40 | 32.54 | |
LibAddressSet.sol | 100 | 100 | 100 | 100 | |
LibConstants.sol | 100 | 100 | 100 | 100 | |
LibDiamond.sol | 1.35 | 0 | 7.69 | 2.3 |... 201,202,205 |
LibHelpers.sol | 100 | 100 | 100 | 100 | |
LibPaginate.sol | 100 | 87.5 | 100 | 100 | |
marketplace/ | 100 | 94.12 | 100 | 100 | |
MarketplaceAccessFacet.sol | 100 | 100 | 100 | 100 | |
MarketplaceInitFacet.sol | 100 | 75 | 100 | 100 | |
MarketplaceTokenHoldersFacet.sol | 100 | 87.5 | 100 | 100 | |
MarketplaceTopFacet.sol | 100 | 100 | 100 | 100 | |
marketplace/lib/ | 100 | 83.33 | 100 | 91.67 | |
AMarketplaceFacet.sol | 100 | 83.33 | 100 | 88.89 | 27 |
IMarketplaceEvents.sol | 100 | 100 | 100 | 100 | |
LibMarketplace.sol | 100 | 100 | 100 | 100 | |
LibMarketplaceAccess.sol | 100 | 100 | 100 | 100 | |
LibMarketplaceTokenHolders.sol | 100 | 100 | 100 | 100 | |

Conclusion:

The code coverage results were obtained by running npx hardhat coverage. We found the :

- Statements Coverage : 79.67%
- Branches Coverage : 80.47%
- Functions Coverage : 71.59%
- Lines Coverage : 89.71%

/

79.67% Statements 294/369 **71.59%** Branches 189/264 **89.71%** Functions 122/136 **80.47%** Lines 408/507

File		Statements		Branches		Functions		Lines	
fast/		100%	182/182	91.07%	102/112	100%	62/62	98.68%	225/228
fast/lib/		88.89%	8/9	77.78%	14/18	92.31%	12/13	83.87%	26/31
interfaces/		100%	0/0	100%	0/0	100%	0/0	100%	0/0
issuer/		100%	34/34	88.89%	16/18	100%	15/15	97.96%	48/49
issuer/lib/		66.67%	2/3	50%	3/6	80%	4/5	63.64%	7/11
lib/		29.13%	30/103	24.29%	17/70	40%	8/20	32.54%	41/126
marketplace/		100%	35/35	94.12%	32/34	100%	15/15	100%	50/50
marketplace/lib/		100%	3/3	83.33%	5/6	100%	6/6	91.67%	11/12

8 Static Analysis (Slither)

Description:

ShellBoxes expanded the coverage of the specific contract areas using automated testing methodologies. Slither, a Solidity static analysis framework, was one of the tools used. Slither was run on all-scoped contracts in both text and binary formats. This tool can be used to test mathematical relationships between Solidity instances statically and variables that allow for the detection of errors or inconsistent usage of the contracts' APIs throughout the entire codebase.

Results:

```
'npx hardhat compile --force' running
Generating typings for: 45 artifacts in dir: typechain for target:
    ↪ ethers-v5
Successfully generated 90 typings!
Compiled 45 Solidity files successfully

LibDiamond.diamondCut(IDiamondCut.FacetCut[],address,bytes).facetIndex (
    ↪ contracts/lib/LibDiamond.sol#70) is a local variable never
    ↪ initialized
LibDiamond.addFunctions(address,bytes4[]).selectorIndex (contracts/lib/
    ↪ LibDiamond.sol#95) is a local variable never initialized
LibDiamond.removeFunctions(address,bytes4[]).selectorIndex (contracts/
    ↪ lib/LibDiamond.sol#128) is a local variable never initialized
LibDiamond.replaceFunctions(address,bytes4[]).selectorIndex (contracts/
    ↪ lib/LibDiamond.sol#113) is a local variable never initialized
Reference: https://github.com/crytic/slither/wiki/Detector-Documentation
    ↪ #uninitialized-local-variables
FastInitFacet.initialize(FastInitFacet.InitializerParams) (contracts/
    ↪ fast/FastInitFacet.sol#42-90) ignores return value by
    ↪ ICustomErrors.AlreadyInitialized() (contracts/fast/FastInitFacet.
    ↪ sol#47)
```

```
FastTokenFacet.mint(uint256,string) (contracts/fast/FastTokenFacet.sol
    ↪ #41-61) ignores return value by ICustomErrors.
    ↪ RequiresContinuousSupply() (contracts/fast/FastTokenFacet.sol#49)
FastTokenFacet.burn(uint256,string) (contracts/fast/FastTokenFacet.sol
    ↪ #79-97) ignores return value by ICustomErrors.
    ↪ RequiresContinuousSupply() (contracts/fast/FastTokenFacet.sol#85)
FastTokenFacet.allowance(address,address) (contracts/fast/FastTokenFacet
    ↪ .sol#276-288) ignores return value by ICustomErrors.
    ↪ RequiresFastGovernorship(spender) (contracts/fast/FastTokenFacet.
    ↪ sol#283)
FastTokenFacet.performTransfer(FastTokenFacet.TransferArgs) (contracts/
    ↪ fast/FastTokenFacet.sol#416-486) ignores return value by
    ↪ ICustomErrors.UnsupportedOperation() (contracts/fast/
    ↪ FastTokenFacet.sol#428)
FastTokenFacet.performTransfer(FastTokenFacet.TransferArgs) (contracts/
    ↪ fast/FastTokenFacet.sol#416-486) ignores return value by
    ↪ ICustomErrors.RequiresFastGovernorship(p.spender) (contracts/fast
    ↪ /FastTokenFacet.sol#432)
FastTokenFacet.beforeRemovingMember(address) (contracts/fast/
    ↪ FastTokenFacet.sol#560-587) ignores return value by ICustomErrors
    ↪ .RequiresPositiveBalance(member) (contracts/fast/FastTokenFacet.
    ↪ sol#563)
FastTokenFacet.onlyTokenHolder(address) (contracts/fast/FastTokenFacet.
    ↪ sol#628-645) ignores return value by ICustomErrors.
    ↪ RequiresMarketplaceMembership(candidate) (contracts/fast/
    ↪ FastTokenFacet.sol#634)
FastTokenFacet.onlyTokenHolder(address) (contracts/fast/FastTokenFacet.
    ↪ sol#628-645) ignores return value by ICustomErrors.
    ↪ RequiresFastMembership(candidate) (contracts/fast/FastTokenFacet.
    ↪ sol#640)
FastTopFacet.setIsSemiPublic(bool) (contracts/fast/FastTopFacet.sol
    ↪ #57-67) ignores return value by ICustomErrors.
    ↪ UnsupportedOperation() (contracts/fast/FastTopFacet.sol#62)
```

```
AFastFacet.onlyDiamondFacet() (contracts/fast/lib/AFastFacet.sol#27-32)
    ↪ ignores return value by ICustomErrors.InternalMethod() (contracts
    ↪ /fast/lib/AFastFacet.sol#29)
AFastFacet.onlyDiamondOwner() (contracts/fast/lib/AFastFacet.sol#35-40)
    ↪ ignores return value by ICustomErrors.RequiresDiamondOwnership()
    ↪ (contracts/fast/lib/AFastFacet.sol#37)
AFastFacet.onlyDeployer() (contracts/fast/lib/AFastFacet.sol#43-48)
    ↪ ignores return value by ICustomErrors.InternalMethod() (contracts
    ↪ /fast/lib/AFastFacet.sol#45)
AFastFacet.onlyMarketplaceMember(address) (contracts/fast/lib/AFastFacet
    ↪ .sol#54-59) ignores return value by ICustomErrors.
    ↪ RequiresMarketplaceMembership(candidate) (contracts/fast/lib/
    ↪ AFastFacet.sol#56)
AFastFacet.onlyMarketplaceActiveMember(address) (contracts/fast/lib/
    ↪ AFastFacet.sol#65-70) ignores return value by ICustomErrors.
    ↪ RequiresMarketplaceActiveMember(candidate) (contracts/fast/lib/
    ↪ AFastFacet.sol#67)
AFastFacet.onlyIssuerMember() (contracts/fast/lib/AFastFacet.sol#75-80)
    ↪ ignores return value by ICustomErrors.RequiresIssuerMembership(
    ↪ msg.sender) (contracts/fast/lib/AFastFacet.sol#77)
AFastFacet.onlyGovernor(address) (contracts/fast/lib/AFastFacet.sol
    ↪ #86-91) ignores return value by ICustomErrors.
    ↪ RequiresFastGovernorship(candidate) (contracts/fast/lib/
    ↪ AFastFacet.sol#88)
AFastFacet.onlyMember(address) (contracts/fast/lib/AFastFacet.sol
    ↪ #97-102) ignores return value by ICustomErrors.
    ↪ RequiresFastMembership(candidate) (contracts/fast/lib/AFastFacet.
    ↪ sol#99)
AFastFacet.differentAddresses(address,address) (contracts/fast/lib/
    ↪ AFastFacet.sol#109-114) ignores return value by ICustomErrors.
    ↪ RequiresDifferentSenderAndRecipient(a) (contracts/fast/lib/
    ↪ AFastFacet.sol#111)
IssuerAccessFacet.removeMember(address) (contracts/issuer/
    ↪ IssuerAccessFacet.sol#73-84) ignores return value by
```

```
    ↪ ICustomErrors.CannotSelfRemove(msg.sender) (contracts/issuer/
    ↪ IssuerAccessFacet.sol#78)
IssuerAccessFacet.governorAddedToFast(address) (contracts/issuer/
    ↪ IssuerAccessFacet.sol#89-99) ignores return value by
    ↪ ICustomErrors.RequiresFastContractCaller() (contracts/issuer/
    ↪ IssuerAccessFacet.sol#93)
IssuerAccessFacet.governorRemovedFromFast(address) (contracts/issuer/
    ↪ IssuerAccessFacet.sol#104-114) ignores return value by
    ↪ ICustomErrors.RequiresFastContractCaller() (contracts/issuer/
    ↪ IssuerAccessFacet.sol#108)
IssuerInitFacet.initialize(IssuerInitFacet.InitializerParams) (contracts
    ↪ /issuer/IssuerInitFacet.sol#33-63) ignores return value by
    ↪ ICustomErrors.AlreadyInitialized() (contracts/issuer/
    ↪ IssuerInitFacet.sol#39)
IssuerTopFacet.registerFast(address) (contracts/issuer/IssuerTopFacet.
    ↪ sol#44-60) ignores return value by ICustomErrors.DuplicateEntry()
    ↪  (contracts/issuer/IssuerTopFacet.sol#50)
AIssuerFacet.onlyDiamondFacet() (contracts/issuer/lib/AIssuerFacet.sol
    ↪ #23-28) ignores return value by ICustomErrors.InternalMethod() (
    ↪ contracts/issuer/lib/AIssuerFacet.sol#25)
AIssuerFacet.onlyDiamondOwner() (contracts/issuer/lib/AIssuerFacet.sol
    ↪ #31-36) ignores return value by ICustomErrors.
    ↪ RequiresDiamondOwnership() (contracts/issuer/lib/AIssuerFacet.sol
    ↪ #33)
AIssuerFacet.onlyMember(address) (contracts/issuer/lib/AIssuerFacet.sol
    ↪ #39-44) ignores return value by ICustomErrors.
    ↪ RequiresIssuerMembership(candidate) (contracts/issuer/lib/
    ↪ AIssuerFacet.sol#41)
MarketplaceAccessFacet.removeMember(address) (contracts/marketplace/
    ↪ MarketplaceAccessFacet.sol#75-87) ignores return value by
    ↪ ICustomErrors.RequiresNoFastMemberships(member) (contracts/
    ↪ marketplace/MarketplaceAccessFacet.sol#81)
MarketplaceAccessFacet.memberAddedToFast(address) (contracts/marketplace
    ↪ /MarketplaceAccessFacet.sol#104-112) ignores return value by
```

```
    ↪ ICustomErrors.RequiresFastContractCaller() (contracts/marketplace
    ↪ /MarketplaceAccessFacet.sol#108)
MarketplaceAccessFacet.memberRemovedFromFast(address) (contracts/
    ↪ marketplace/MarketplaceAccessFacet.sol#118-125) ignores return
    ↪ value by ICustomErrors.RequiresFastContractCaller() (contracts/
    ↪ marketplace/MarketplaceAccessFacet.sol#121)
MarketplaceAccessFacet.activateMember(address) (contracts/marketplace/
    ↪ MarketplaceAccessFacet.sol#139-154) ignores return value by
    ↪ ICustomErrors.RequiresMarketplaceDeactivatedMember(member) (
    ↪ contracts/marketplace/MarketplaceAccessFacet.sol#146)
MarketplaceAccessFacet.deactivateMember(address) (contracts/marketplace/
    ↪ MarketplaceAccessFacet.sol#160-175) ignores return value by
    ↪ ICustomErrors.RequiresMarketplaceActiveMember(member) (contracts/
    ↪ marketplace/MarketplaceAccessFacet.sol#167)
MarketplaceInitFacet.initialize(MarketplaceInitFacet.InitializerParams)
    ↪ (contracts/marketplace/MarketplaceInitFacet.sol#20-52) ignores
    ↪ return value by ICustomErrors.AlreadyInitialized() (contracts/
    ↪ marketplace/MarketplaceInitFacet.sol#25)
MarketplaceTokenHoldersFacet.fastBalanceChanged(address,uint256) (
    ↪ contracts/marketplace/MarketplaceTokenHoldersFacet.sol#16-39)
    ↪ ignores return value by ICustomErrors.RequiresFastContractCaller
    ↪ () (contracts/marketplace/MarketplaceTokenHoldersFacet.sol#25)
AMarketplaceFacet.onlyDeployer() (contracts/marketplace/lib/
    ↪ AMarketplaceFacet.sol#25-30) ignores return value by
    ↪ ICustomErrors.InternalMethod() (contracts/marketplace/lib/
    ↪ AMarketplaceFacet.sol#27)
AMarketplaceFacet.onlyIssuerMember() (contracts/marketplace/lib/
    ↪ AMarketplaceFacet.sol#35-40) ignores return value by
    ↪ ICustomErrors.RequiresIssuerMembership(msg.sender) (contracts/
    ↪ marketplace/lib/AMarketplaceFacet.sol#37)
AMarketplaceFacet.onlyMember(address) (contracts/marketplace/lib/
    ↪ AMarketplaceFacet.sol#46-51) ignores return value by
    ↪ ICustomErrors.RequiresMarketplaceMembership(candidate) (contracts
    ↪ /marketplace/lib/AMarketplaceFacet.sol#48)
```

```
Reference: https://github.com/crytic/slither/wiki/Detector-Documentation
    ↪ #unused-return
FastTokenFacet.beforeRemovingMember(address) (contracts/fast/
    ↪ FastTokenFacet.sol#560-587) has external calls inside a loop:
    ↪ this.performDisapproval(member,spender,s.allowances[member][
    ↪ spender]) (contracts/fast/FastTokenFacet.sol#574)
FastTokenFacet.beforeRemovingMember(address) (contracts/fast/
    ↪ FastTokenFacet.sol#560-587) has external calls inside a loop:
    ↪ this.performDisapproval(owner,member,s.allowances[owner][member])
    ↪  (contracts/fast/FastTokenFacet.sol#584)
IssuerFrontendFacet.paginateDetailedFasts(uint256,uint256) (contracts/
    ↪ issuer/IssuerFrontendFacet.sol#22-33) has external calls inside a
    ↪  loop: fastDetails[i] = FastFrontendFacet(addresses[i]).details()
    ↪  (contracts/issuer/IssuerFrontendFacet.sol#29)
Reference: https://github.com/crytic/slither/wiki/Detector-Documentation
    ↪ /#calls-inside-a-loop
LibFast.data() (contracts/fast/lib/LibFast.sol#40-43) uses assembly
        - INLINE ASM (contracts/fast/lib/LibFast.sol#42)
LibFastAccess.data() (contracts/fast/lib/LibFastAccess.sol#35-38) uses
    ↪ assembly
        - INLINE ASM (contracts/fast/lib/LibFastAccess.sol#37)
LibFastHistory.data() (contracts/fast/lib/LibFastHistory.sol#72-75) uses
    ↪  assembly
        - INLINE ASM (contracts/fast/lib/LibFastHistory.sol#74)
LibFastToken.data() (contracts/fast/lib/LibFastToken.sol#63-66) uses
    ↪ assembly
        - INLINE ASM (contracts/fast/lib/LibFastToken.sol#65)
LibIssuer.data() (contracts/issuer/lib/LibIssuer.sol#24-27) uses
    ↪ assembly
        - INLINE ASM (contracts/issuer/lib/LibIssuer.sol#26)
LibIssuerAccess.data() (contracts/issuer/lib/LibIssuerAccess.sol#22-25)
    ↪ uses assembly
        - INLINE ASM (contracts/issuer/lib/LibIssuerAccess.sol#24)
```

```
LibDiamond.diamondStorage() (contracts/lib/LibDiamond.sol#38-43) uses
    ↪ assembly
        - INLINE ASM (contracts/lib/LibDiamond.sol#40-42)
LibDiamond.enforceHasContractCode(address,string) (contracts/lib/
    ↪ LibDiamond.sol#200-206) uses assembly
        - INLINE ASM (contracts/lib/LibDiamond.sol#202-204)
LibMarketplace.data() (contracts/marketplace/lib/LibMarketplace.sol
    ↪ #20-23) uses assembly
        - INLINE ASM (contracts/marketplace/lib/LibMarketplace.sol#22)
LibMarketplaceAccess.data() (contracts/marketplace/lib/
    ↪ LibMarketplaceAccess.sol#24-27) uses assembly
        - INLINE ASM (contracts/marketplace/lib/LibMarketplaceAccess.sol
            ↪ #26)
LibMarketplaceTokenHolders.data() (contracts/marketplace/lib/
    ↪ LibMarketplaceTokenHolders.sol#20-23) uses assembly
        - INLINE ASM (contracts/marketplace/lib/
            ↪ LibMarketplaceTokenHolders.sol#22)
Reference: https://github.com/crytic/slither/wiki/Detector-Documentation
    ↪ #assembly-usage
Pragma version0.8.10 (contracts/fast/FastAccessFacet.sol#2) necessitates
    ↪  a version too recent to be trusted. Consider deploying with
    ↪ 0.6.12/0.7.6/0.8.7
Pragma version0.8.10 (contracts/fast/FastFrontendFacet.sol#2)
    ↪ necessitates a version too recent to be trusted. Consider
    ↪ deploying with 0.6.12/0.7.6/0.8.7
Pragma version0.8.10 (contracts/fast/FastHistoryFacet.sol#2)
    ↪ necessitates a version too recent to be trusted. Consider
    ↪ deploying with 0.6.12/0.7.6/0.8.7
Pragma version0.8.10 (contracts/fast/FastInitFacet.sol#2) necessitates a
    ↪  version too recent to be trusted. Consider deploying with
    ↪ 0.6.12/0.7.6/0.8.7
Pragma version0.8.10 (contracts/fast/FastTokenFacet.sol#2) necessitates
    ↪ a version too recent to be trusted. Consider deploying with
    ↪ 0.6.12/0.7.6/0.8.7
```

```
Pragma version0.8.10 (contracts/fast/FastTopFacet.sol#2) necessitates a
    ↪ version too recent to be trusted. Consider deploying with
    ↪ 0.6.12/0.7.6/0.8.7
Pragma version0.8.10 (contracts/fast/lib/AFastFacet.sol#2) necessitates
    ↪ a version too recent to be trusted. Consider deploying with
    ↪ 0.6.12/0.7.6/0.8.7
Pragma version0.8.10 (contracts/fast/lib/IFast.sol#2) necessitates a
    ↪ version too recent to be trusted. Consider deploying with
    ↪ 0.6.12/0.7.6/0.8.7
Pragma version0.8.10 (contracts/fast/lib/IFastEvents.sol#2) necessitates
    ↪  a version too recent to be trusted. Consider deploying with
    ↪ 0.6.12/0.7.6/0.8.7
Pragma version0.8.10 (contracts/fast/lib/LibFast.sol#2) necessitates a
    ↪ version too recent to be trusted. Consider deploying with
    ↪ 0.6.12/0.7.6/0.8.7
Pragma version0.8.10 (contracts/fast/lib/LibFastAccess.sol#2)
    ↪ necessitates a version too recent to be trusted. Consider
    ↪ deploying with 0.6.12/0.7.6/0.8.7
Pragma version0.8.10 (contracts/fast/lib/LibFastHistory.sol#2)
    ↪ necessitates a version too recent to be trusted. Consider
    ↪ deploying with 0.6.12/0.7.6/0.8.7
Pragma version0.8.10 (contracts/fast/lib/LibFastToken.sol#2)
    ↪ necessitates a version too recent to be trusted. Consider
    ↪ deploying with 0.6.12/0.7.6/0.8.7
Pragma version0.8.10 (contracts/interfaces/ICustomErrors.sol#2)
    ↪ necessitates a version too recent to be trusted. Consider
    ↪ deploying with 0.6.12/0.7.6/0.8.7
Pragma version0.8.10 (contracts/interfaces/IDiamondCut.sol#2)
    ↪ necessitates a version too recent to be trusted. Consider
    ↪ deploying with 0.6.12/0.7.6/0.8.7
Pragma version0.8.10 (contracts/interfaces/IDiamondLoupe.sol#2)
    ↪ necessitates a version too recent to be trusted. Consider
    ↪ deploying with 0.6.12/0.7.6/0.8.7
```

```
Pragma version0.8.10 (contracts/interfaces/IERC1404.sol#2) necessitates
    ↪ a version too recent to be trusted. Consider deploying with
    ↪ 0.6.12/0.7.6/0.8.7
Pragma version0.8.10 (contracts/interfaces/IERC165.sol#2) necessitates a
    ↪  version too recent to be trusted. Consider deploying with
    ↪ 0.6.12/0.7.6/0.8.7
Pragma version0.8.10 (contracts/interfaces/IERC173.sol#2) necessitates a
    ↪  version too recent to be trusted. Consider deploying with
    ↪ 0.6.12/0.7.6/0.8.7
Pragma version0.8.10 (contracts/interfaces/IERC20.sol#2) necessitates a
    ↪ version too recent to be trusted. Consider deploying with
    ↪ 0.6.12/0.7.6/0.8.7
Pragma version0.8.10 (contracts/interfaces/IHasActiveMembers.sol#2)
    ↪ necessitates a version too recent to be trusted. Consider
    ↪ deploying with 0.6.12/0.7.6/0.8.7
Pragma version0.8.10 (contracts/interfaces/IHasGovernors.sol#2)
    ↪ necessitates a version too recent to be trusted. Consider
    ↪ deploying with 0.6.12/0.7.6/0.8.7
Pragma version0.8.10 (contracts/interfaces/IHasMembers.sol#2)
    ↪ necessitates a version too recent to be trusted. Consider
    ↪ deploying with 0.6.12/0.7.6/0.8.7
Pragma version0.8.10 (contracts/issuer/IssuerAccessFacet.sol#2)
    ↪ necessitates a version too recent to be trusted. Consider
    ↪ deploying with 0.6.12/0.7.6/0.8.7
Pragma version0.8.10 (contracts/issuer/IssuerFrontendFacet.sol#2)
    ↪ necessitates a version too recent to be trusted. Consider
    ↪ deploying with 0.6.12/0.7.6/0.8.7
Pragma version0.8.10 (contracts/issuer/IssuerInitFacet.sol#2)
    ↪ necessitates a version too recent to be trusted. Consider
    ↪ deploying with 0.6.12/0.7.6/0.8.7
Pragma version0.8.10 (contracts/issuer/IssuerTopFacet.sol#2)
    ↪ necessitates a version too recent to be trusted. Consider
    ↪ deploying with 0.6.12/0.7.6/0.8.7
```

```
Pragma version0.8.10 (contracts/issuer/lib/AIssuerFacet.sol#2)
   ↪ necessitates a version too recent to be trusted. Consider
   ↪ deploying with 0.6.12/0.7.6/0.8.7
Pragma version0.8.10 (contracts/issuer/lib/IIssuerEvents.sol#2)
   ↪ necessitates a version too recent to be trusted. Consider
   ↪ deploying with 0.6.12/0.7.6/0.8.7
Pragma version0.8.10 (contracts/issuer/lib/LibIssuer.sol#2) necessitates
   ↪  a version too recent to be trusted. Consider deploying with
   ↪ 0.6.12/0.7.6/0.8.7
Pragma version0.8.10 (contracts/issuer/lib/LibIssuerAccess.sol#2)
   ↪ necessitates a version too recent to be trusted. Consider
   ↪ deploying with 0.6.12/0.7.6/0.8.7
Pragma version0.8.10 (contracts/lib/LibAddressSet.sol#2) necessitates a
   ↪ version too recent to be trusted. Consider deploying with
   ↪ 0.6.12/0.7.6/0.8.7
Pragma version0.8.10 (contracts/lib/LibConstants.sol#2) necessitates a
   ↪ version too recent to be trusted. Consider deploying with
   ↪ 0.6.12/0.7.6/0.8.7
Pragma version0.8.10 (contracts/lib/LibDiamond.sol#2) necessitates a
   ↪ version too recent to be trusted. Consider deploying with
   ↪ 0.6.12/0.7.6/0.8.7
Pragma version0.8.10 (contracts/lib/LibHelpers.sol#2) necessitates a
   ↪ version too recent to be trusted. Consider deploying with
   ↪ 0.6.12/0.7.6/0.8.7
Pragma version0.8.10 (contracts/lib/LibPaginate.sol#2) necessitates a
   ↪ version too recent to be trusted. Consider deploying with
   ↪ 0.6.12/0.7.6/0.8.7
Pragma version0.8.10 (contracts/marketplace/MarketplaceAccessFacet.sol
   ↪ #2) necessitates a version too recent to be trusted. Consider
   ↪ deploying with 0.6.12/0.7.6/0.8.7
Pragma version0.8.10 (contracts/marketplace/MarketplaceInitFacet.sol#2)
   ↪ necessitates a version too recent to be trusted. Consider
   ↪ deploying with 0.6.12/0.7.6/0.8.7
```

```
Pragma version0.8.10 (contracts/marketplace/MarketplaceTokenHoldersFacet
    ↪ .sol#2) necessitates a version too recent to be trusted. Consider
    ↪  deploying with 0.6.12/0.7.6/0.8.7
Pragma version0.8.10 (contracts/marketplace/MarketplaceTopFacet.sol#2)
    ↪ necessitates a version too recent to be trusted. Consider
    ↪ deploying with 0.6.12/0.7.6/0.8.7
Pragma version0.8.10 (contracts/marketplace/lib/AMarketplaceFacet.sol#2)
    ↪  necessitates a version too recent to be trusted. Consider
    ↪ deploying with 0.6.12/0.7.6/0.8.7
Pragma version0.8.10 (contracts/marketplace/lib/IMarketplaceEvents.sol
    ↪ #2) necessitates a version too recent to be trusted. Consider
    ↪ deploying with 0.6.12/0.7.6/0.8.7
Pragma version0.8.10 (contracts/marketplace/lib/LibMarketplace.sol#2)
    ↪ necessitates a version too recent to be trusted. Consider
    ↪ deploying with 0.6.12/0.7.6/0.8.7
Pragma version0.8.10 (contracts/marketplace/lib/LibMarketplaceAccess.sol
    ↪ #2) necessitates a version too recent to be trusted. Consider
    ↪ deploying with 0.6.12/0.7.6/0.8.7
Pragma version0.8.10 (contracts/marketplace/lib/
    ↪ LibMarketplaceTokenHolders.sol#2) necessitates a version too
    ↪ recent to be trusted. Consider deploying with 0.6.12/0.7.6/0.8.7
solc-0.8.10 is not recommended for deployment
Reference: https://github.com/crytic/slither/wiki/Detector-Documentation
    ↪ #incorrect-versions-of-solidity
Low level call in LibDiamond.initializeDiamondCut(address,bytes) (
    ↪ contracts/lib/LibDiamond.sol#180-198):
        - (success,error) = _init.delegatecall(_calldata) (contracts/lib/
            ↪ LibDiamond.sol#188)
Reference: https://github.com/crytic/slither/wiki/Detector-Documentation
    ↪ #low-level-calls
FastTopFacet (contracts/fast/FastTopFacet.sol#11-68) should inherit from
    ↪  IFast (contracts/fast/lib/IFast.sol#6-18)
Reference: https://github.com/crytic/slither/wiki/Detector-Documentation
    ↪ #missing-inheritance
```

```
Parameter LibDiamond.setContractOwner(address)._newOwner (contracts/lib/
    ↪ LibDiamond.sol#47) is not in mixedCase
Parameter LibDiamond.diamondCut(IDiamondCut.FacetCut[],address,bytes).
    ↪ _diamondCut (contracts/lib/LibDiamond.sol#66) is not in mixedCase
Parameter LibDiamond.diamondCut(IDiamondCut.FacetCut[],address,bytes).
    ↪ _init (contracts/lib/LibDiamond.sol#67) is not in mixedCase
Parameter LibDiamond.diamondCut(IDiamondCut.FacetCut[],address,bytes).
    ↪ _calldata (contracts/lib/LibDiamond.sol#68) is not in mixedCase
Parameter LibDiamond.addFunctions(address,bytes4[])._facetAddress (
    ↪ contracts/lib/LibDiamond.sol#86) is not in mixedCase
Parameter LibDiamond.addFunctions(address,bytes4[])._functionSelectors (
    ↪ contracts/lib/LibDiamond.sol#86) is not in mixedCase
Parameter LibDiamond.replaceFunctions(address,bytes4[])._facetAddress (
    ↪ contracts/lib/LibDiamond.sol#104) is not in mixedCase
Parameter LibDiamond.replaceFunctions(address,bytes4[]).
    ↪ _functionSelectors (contracts/lib/LibDiamond.sol#104) is not in
    ↪ mixedCase
Parameter LibDiamond.removeFunctions(address,bytes4[])._facetAddress (
    ↪ contracts/lib/LibDiamond.sol#123) is not in mixedCase
Parameter LibDiamond.removeFunctions(address,bytes4[]).
    ↪ _functionSelectors (contracts/lib/LibDiamond.sol#123) is not in
    ↪ mixedCase
Parameter LibDiamond.addFacet(LibDiamond.DiamondStorage,address).
    ↪ _facetAddress (contracts/lib/LibDiamond.sol#135) is not in
    ↪ mixedCase
Parameter LibDiamond.addFunction(LibDiamond.DiamondStorage,bytes4,uint96
    ↪ ,address)._selector (contracts/lib/LibDiamond.sol#142) is not in
    ↪ mixedCase
Parameter LibDiamond.addFunction(LibDiamond.DiamondStorage,bytes4,uint96
    ↪ ,address)._selectorPosition (contracts/lib/LibDiamond.sol#142) is
    ↪  not in mixedCase
Parameter LibDiamond.addFunction(LibDiamond.DiamondStorage,bytes4,uint96
    ↪ ,address)._facetAddress (contracts/lib/LibDiamond.sol#142) is not
    ↪  in mixedCase
```

```
Parameter LibDiamond.removeFunction(LibDiamond.DiamondStorage,address,
    ↪ bytes4)._facetAddress (contracts/lib/LibDiamond.sol#148) is not
    ↪ in mixedCase
Parameter LibDiamond.removeFunction(LibDiamond.DiamondStorage,address,
    ↪ bytes4)._selector (contracts/lib/LibDiamond.sol#148) is not in
    ↪ mixedCase
Parameter LibDiamond.initializeDiamondCut(address,bytes)._init (
    ↪ contracts/lib/LibDiamond.sol#180) is not in mixedCase
Parameter LibDiamond.initializeDiamondCut(address,bytes)._calldata (
    ↪ contracts/lib/LibDiamond.sol#180) is not in mixedCase
Parameter LibDiamond.enforceHasContractCode(address,string)._contract (
    ↪ contracts/lib/LibDiamond.sol#200) is not in mixedCase
Parameter LibDiamond.enforceHasContractCode(address,string).
    ↪ _errorMessage (contracts/lib/LibDiamond.sol#200) is not in
    ↪ mixedCase
Reference: https://github.com/crytic/slither/wiki/Detector-Documentation
    ↪ #conformance-to-solidity-naming-conventions
LibFast.STORAGE_VERSION (contracts/fast/lib/LibFast.sol#13) is never
    ↪ used in LibFast (contracts/fast/lib/LibFast.sol#11-44)
LibFastAccess.STORAGE_VERSION (contracts/fast/lib/LibFastAccess.sol#14)
    ↪ is never used in LibFastAccess (contracts/fast/lib/LibFastAccess.
    ↪ sol#12-39)
LibFastHistory.STORAGE_VERSION (contracts/fast/lib/LibFastHistory.sol
    ↪ #14) is never used in LibFastHistory (contracts/fast/lib/
    ↪ LibFastHistory.sol#12-76)
LibFastToken.STORAGE_VERSION (contracts/fast/lib/LibFastToken.sol#15) is
    ↪  never used in LibFastToken (contracts/fast/lib/LibFastToken.sol
    ↪ #13-67)
LibFastToken.DEFAULT_TRANSFER_REFERENCE (contracts/fast/lib/LibFastToken
    ↪ .sol#21) is never used in LibFastToken (contracts/fast/lib/
    ↪ LibFastToken.sol#13-67)
LibIssuer.STORAGE_VERSION (contracts/issuer/lib/LibIssuer.sol#9) is
    ↪ never used in LibIssuer (contracts/issuer/lib/LibIssuer.sol#7-28)
```

```
LibIssuerAccess.STORAGE_VERSION (contracts/issuer/lib/LibIssuerAccess.
    ↪ sol#9) is never used in LibIssuerAccess (contracts/issuer/lib/
    ↪ LibIssuerAccess.sol#7-26)
LibMarketplace.STORAGE_VERSION (contracts/marketplace/lib/LibMarketplace
    ↪ .sol#9) is never used in LibMarketplace (contracts/marketplace/
    ↪ lib/LibMarketplace.sol#7-24)
LibMarketplaceAccess.STORAGE_VERSION (contracts/marketplace/lib/
    ↪ LibMarketplaceAccess.sol#9) is never used in LibMarketplaceAccess
    ↪  (contracts/marketplace/lib/LibMarketplaceAccess.sol#7-28)
LibMarketplaceTokenHolders.STORAGE_VERSION (contracts/marketplace/lib/
    ↪ LibMarketplaceTokenHolders.sol#9) is never used in
    ↪ LibMarketplaceTokenHolders (contracts/marketplace/lib/
    ↪ LibMarketplaceTokenHolders.sol#7-24)
Reference: https://github.com/crytic/slither/wiki/Detector-Documentation
    ↪ #unused-state-variable
details() should be declared external:
        - FastFrontendFacet.details() (contracts/fast/FastFrontendFacet.
            ↪ sol#107-126)
allowance(address,address) should be declared external:
        - FastTokenFacet.allowance(address,address) (contracts/fast/
            ↪ FastTokenFacet.sol#276-288)
Reference: https://github.com/crytic/slither/wiki/Detector-Documentation
    ↪ #public-function-that-could-be-declared-external
. analyzed (45 contracts with 72 detectors), 135 result(s) found
```

Conclusion:

Most of the vulnerabilities found by the analysis have already been addressed by the smart contract code review.

9 Conclusion

In this audit, we examined the design and implementation of Consilience contract and discovered several issues of varying severity. Consilience Group Limited team addressed 2 issues raised in the initial report and implemented the necessary fixes, while acknowledging the rest due to legal and business logic requirements.

10 Disclaimer

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